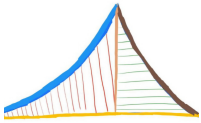
Math for EE



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About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. CBSE 2019

A. Formula-12

1.1.1 Multiplicative Property of Determinants. For any two square matrices $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n\times n}$:

$$\left|\mathbf{AB}\right| = \left|\mathbf{A}\right|\left|\mathbf{B}\right| \tag{1.1.1.1}$$

For an order-2 square matrix $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$:

$$\left|\mathbf{A}\right| = ad - bc \tag{1.1.1.2}$$

1.1.2 Derivative of an Exponential Function via Chain Rule:

$$\frac{d}{dx}\left[e^{u(x)}\right] = e^{u(x)}\frac{du}{dx} \tag{1.1.2.1}$$

For $u(x) = kx$:

$$\frac{d}{dx}\left[e^{kx}\right] = ke^{kx} \tag{1.1.2.2}$$

1.1.3 Order and Degree of an Ordinary Differential Equation. For a differential equation relating x, y and derivatives $y^{(k)} = \frac{d^k y}{dx^k}$:

- **Order:** The highest order n of the derivative $\dfrac{d^n y}{dx^n}$ present in the equation.
- **Degree:** The highest power/exponent of the highest-order derivative, defined *only* if the equation can be expressed as a polynomial in all its derivatives. If any derivative occurs within transcendental functions (such as $\log$, $\sin$, e^x), the degree is **not defined**.

1.1.4 Direction Cosines and Direction Ratios of a Line. If a line makes angles α, β, γ with the positive x, y, z axes:

$$l = \cos\alpha, \quad m = \cos\beta, \quad n = \cos\gamma \tag{1.1.4.1}$$

The fundamental identity connecting direction cosines is:

$$l^2 + m^2 + n^2 = 1 \tag{1.1.4.2}$$

If the line makes equal angles with all three coordinate axes ($\alpha = \beta = \gamma$):

$$l = m = n = \pm\frac{1}{\sqrt{3}} \tag{1.1.4.3}$$

1.1.5 Vector and Cartesian Equations of a Straight Line in 3D. A line passing through a point with position vector $\mathbf{a} = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$ parallel to direction vector $\mathbf{b} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ has:

$$\text{Vector Form: } \mathbf{r} = \mathbf{a} + \lambda\mathbf{b}, \quad \lambda \in \mathbb{R} \tag{1.1.5.1}$$

$$\text{Cartesian Form: } \frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c} = \lambda \tag{1.1.5.2}$$

1.1.6 Power Rule of Indefinite Integration:

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1 \tag{1.1.6.1}$$

1.1.7 Determinant of Powers of a Matrix:

$$\left|\mathbf{A}^k\right| = \left(\left|\mathbf{A}\right|\right)^k, \quad k \in \mathbb{Z}^+ \tag{1.1.7.1}$$

1.1.8 Binary Operations and Associativity. An operation $*$ on a set S is a binary operation if

$$a * b \in S \quad \forall a, b \in S \quad \text{(Closure)} \tag{1.1.8.1}$$

The operation is associative if

$$(a * b) * c = a * (b * c) \quad \forall a, b, c \in S \tag{1.1.8.2}$$

1.1.9 Vector Cross Product and Sine of Angle between Vectors. For non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ with angle $\theta \in [0, \pi]$:

$$\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \, \|\mathbf{b}\| \sin\theta \tag{1.1.9.1}$$

$$\sin\theta = \frac{\|\mathbf{a} \times \mathbf{b}\|}{\|\mathbf{a}\| \, \|\mathbf{b}\|} \tag{1.1.9.2}$$

1.1.10 Scalar Triple Product and Volume of Parallelepiped. The volume of a parallelepiped with coterminous edge vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ is

$$V = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})| = \left|\begin{matrix}\mathbf{a} & \mathbf{b} & \mathbf{c}\end{matrix}\right| \tag{1.1.10.1}$$

1.1.11 Binomial Distribution for Bernoulli Trials. In n independent trials with success probability p and failure probability $q = 1 - p$:

$$\Pr(X = k) = \binom{n}{k} p^k q^{n-k}, \quad k \in \{0, 1, \ldots, n\} \tag{1.1.11.1}$$

The cumulative distribution satisfies:

$$\Pr(X \leq k) = \sum_{r=0}^{k} \binom{n}{r} p^r q^{n-r} = 1 - \Pr(X > k) \tag{1.1.11.2}$$

1.1.12 Discrete Probability Distribution Axioms. For a discrete random variable X taking values $\{x_1, x_2, \ldots, x_k\}$:

$$\Pr(X = x_i) \geq 0 \quad \forall i, \qquad \sum_{i=1}^{k} \Pr(X = x_i) = 1 \tag{1.1.12.1}$$

1.1.13 Standard Trigonometric Integrals:

$$\int \sec x \tan x \, dx = \sec x + C \tag{1.1.13.1}$$

$$\int \csc x \cot x \, dx = -\csc x + C \tag{1.1.13.2}$$

1.1.14 Special Integral Form $\int e^x \left[f(x) + f'(x) \right] dx$:

$$\int e^x \left[f(x) + f'(x) \right] dx = e^x f(x) + C \tag{1.1.14.1}$$

Solution: Applying integration by parts to the first term $\int e^x f(x) \, dx$:

$$\int e^x f(x) \, dx = f(x) \int e^x \, dx - \int \left(f'(x) \int e^x \, dx \right) dx \tag{1.1.14.2}$$

$$= f(x) e^x - \int f'(x) e^x \, dx \tag{1.1.14.3}$$

Adding $\int e^x f'(x) \, dx$ to both sides yields (1.1.14.1).

1.1.15 Conditional Probability and Bayes' Inversion:

$$\Pr(A \mid B) = \frac{\Pr(AB)}{\Pr(B)}, \quad \Pr(B \mid A) = \frac{\Pr(AB)}{\Pr(A)} \tag{1.1.15.1}$$

$$\implies \Pr(A \mid B) = \frac{\Pr(B \mid A) \Pr(A)}{\Pr(B)} \tag{1.1.15.2}$$

1.1.16 Separation of Variables for First-Order ODEs:

$$\frac{dy}{dx} = g(x)h(y) \implies \int \frac{1}{h(y)} \, dy = \int g(x) \, dx + C \tag{1.1.16.1}$$

1.1.17 Inverse Trigonometric Identities:

$$\cos^{-1} x = \tan^{-1} \left(\frac{\sqrt{1 - x^2}}{x} \right), \quad 0 < x \leq 1 \tag{1.1.17.1}$$

$$\tan^{-1} u + \tan^{-1} v = \tan^{-1} \left(\frac{u + v}{1 - uv} \right), \quad uv < 1 \tag{1.1.17.2}$$

$$\sin\left(\tan^{-1} w \right) = \frac{w}{\sqrt{1 + w^2}} \tag{1.1.17.3}$$

1.1.18 Invariance of Determinants under Elementary Row/Column Operations:

$$\left| \mathbf{A} \right| = \left| \mathbf{EA} \right| \quad \text{for shear operations } R_i \to R_i + kR_j \text{ or } C_i \to C_i + kC_j \tag{1.1.18.1}$$

1.1.19 Equivalence Relation Criteria. A binary relation R on a set A is:

- **Reflexive:** If $(a, a) \in R$ for all $a \in A$.
- **Symmetric:** If $(a, b) \in R \implies (b, a) \in R$.

- **Transitive:** If $(a,b) \in R$ and $(b,c) \in R \implies (a,c) \in R$.

1.1.20 Bijective Maps and Functional Invertibility. A function $f : X \to Y$ is invertible if and only if it is a bijection:

$$\text{Injective: } f(x_1) = f(x_2) \implies x_1 = x_2 \quad \forall x_1, x_2 \in X \tag{1.1.20.1}$$

$$\text{Surjective: } \forall y \in Y,\ \exists x \in X \text{ such that } f(x) = y \tag{1.1.20.2}$$

The inverse function satisfies $f^{-1}(y) = x \iff f(x) = y$.

1.1.21 Logarithmic Differentiation. For variable bases and exponents $u(x)^{v(x)} = w(x)^{z(x)}$:

$$v(x) \ln\left(u(x)\right) = z(x) \ln\left(w(x)\right) \tag{1.1.21.1}$$

Differentiating implicitly using the product rule and chain rule:

$$\frac{d}{dx}\left[v \ln u\right] = \frac{dv}{dx} \ln u + \frac{v}{u}\frac{du}{dx} \tag{1.1.21.2}$$

1.1.22 King's Property of Definite Integrals:

$$\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx \tag{1.1.22.1}$$

Solution: Substitute $t = a - x \implies dx = -dt$. When $x = 0$, $t = a$; when $x = a$, $t = 0$:

$$\int_0^a f(x)\,dx = \int_a^0 f(a-t)\,(-dt) = \int_0^a f(a-t)\,dt = \int_0^a f(a-x)\,dx \tag{1.1.22.2}$$

yielding (1.1.22.1). For symmetric intervals of even functions:

$$\int_{-a}^a g(u)\,du = 2\int_0^a g(u)\,du \quad \text{if } g(-u) = g(u) \tag{1.1.22.3}$$

1.1.23 Partial Fraction Decomposition for Irreducible Quadratic Factors:

$$\frac{P(x)}{(x-\alpha)(x^2+\beta^2)} = \frac{A}{x-\alpha} + \frac{Bx+C}{x^2+\beta^2} \tag{1.1.23.1}$$

Standard primitive integrals:

$$\int \frac{1}{x-\alpha}\,dx = \ln|x-\alpha| + C \tag{1.1.23.2}$$

$$\int \frac{2x}{x^2+\beta^2}\,dx = \ln\left(x^2+\beta^2\right) + C \tag{1.1.23.3}$$

$$\int \frac{1}{x^2+\beta^2}\,dx = \frac{1}{\beta}\tan^{-1}\left(\frac{x}{\beta}\right) + C \tag{1.1.23.4}$$

1.1.24 Homogeneous First-Order Differential Equations:

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right) \tag{1.1.24.1}$$

Solution: Substituting $y = vx \implies \frac{dy}{dx} = v + x\frac{dv}{dx}$:

$$v + x\frac{dv}{dx} = F(v) \implies \frac{dv}{F(v) - v} = \frac{dx}{x} \tag{1.1.24.2}$$

reducing the equation to separable variables.

1.1.25 First-Order Linear Differential Equations:

$$\frac{dy}{dx} + P(x)y = Q(x) \tag{1.1.25.1}$$

Solution: Multiplying by the integrating factor $\mu(x) = e^{\int P(x)\,dx}$:

$$\frac{d}{dx}\left[y \cdot \mu(x)\right] = Q(x)\mu(x) \tag{1.1.25.2}$$

$$\implies y \cdot e^{\int P(x)\,dx} = \int Q(x)e^{\int P(x)\,dx}\,dx + C \tag{1.1.25.3}$$

1.1.26 Unit Vector and Orthogonal Vector Projection:

$$\hat{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|}, \quad \|\hat{u}\| = 1 \tag{1.1.26.1}$$

For vectors $\mathbf{a}$ and $\mathbf{b}$:

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z \tag{1.1.26.2}$$

1.1.27 Related Rates of Change via Derivative Chains. If $V(t) = f(x(t))$ and $S(t) = g(x(t))$:

$$\frac{dV}{dt} = f'(x)\frac{dx}{dt}, \quad \frac{dS}{dt} = g'(x)\frac{dx}{dt} \implies \frac{dS}{dt} = \frac{g'(x)}{f'(x)}\frac{dV}{dt} \tag{1.1.27.1}$$

For a cube with side length x:

$$V = x^3 \implies \frac{dV}{dt} = 3x^2\frac{dx}{dt}, \quad S = 6x^2 \implies \frac{dS}{dt} = 12x\frac{dx}{dt} \tag{1.1.27.2}$$

1.1.28 Plane Passing Through Three Non-Collinear Points. The Cartesian equation of a plane through $\mathbf{a} = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$, and $\mathbf{c} = \begin{pmatrix} x_3 \\ y_3 \\ z_3 \end{pmatrix}$ is

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0 \tag{1.1.28.1}$$

In vector form with normal $\mathbf{n} = (\mathbf{b} - \mathbf{a}) \times (\mathbf{c} - \mathbf{a})$:

$$\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n} \tag{1.1.28.2}$$

1.1.29 Calculus-Based Extremum and Optimization. For a twice-differentiable functional

$Z(y)$:

$$\text{Stationary Condition: } \frac{dZ}{dy} = 0 \tag{1.1.29.1}$$

$$\text{Local Minimum: } \left.\frac{d^2Z}{dy^2}\right|_{y=y^*} > 0 \tag{1.1.29.2}$$

1.1.30 Definite Integral as the Limit of Riemann Sums:

$$\int_a^b f(x)\,dx = \lim_{h\to 0} h\sum_{r=1}^{n} f(a+rh), \quad \text{where } nh = b-a \tag{1.1.30.1}$$

Fundamental summation formulas:

$$\sum_{r=1}^{n} r = \frac{n(n+1)}{2} \tag{1.1.30.2}$$

$$\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6} \tag{1.1.30.3}$$

$$\sum_{r=1}^{n} q^r = q\frac{q^n-1}{q-1}, \quad q \neq 1 \tag{1.1.30.4}$$

Asymptotic exponential limit:

$$\lim_{h\to 0} \frac{e^{kh}-1}{h} = k \tag{1.1.30.5}$$

1.1.31 Area of Plane Region Bounded by Curves:

$$A = \int_a^b \left(y_{\text{upper}}(x) - y_{\text{lower}}(x)\right)\,dx \tag{1.1.31.1}$$

1.1.32 Matrix Inverse via Adjugate Matrix:

$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|}\,\text{adj}(\mathbf{A}) = \frac{1}{|\mathbf{A}|}\mathbf{C}^\top \tag{1.1.32.1}$$

where $C_{ij} = (-1)^{i+j}M_{ij}$ is the cofactor of entry a_{ij}. The unique solution to linear system $\mathbf{Ax} = \mathbf{b}$ is:

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} \tag{1.1.32.2}$$

1.1.33 Matrix Inversion by Gauss-Jordan Row Elimination:

$$[\mathbf{A} \mid \mathbf{I}] \xrightarrow{R_i \to cR_i,\, R_i \leftrightarrow R_j,\, R_i \to R_i + kR_j} \left[\mathbf{I} \mid \mathbf{A}^{-1}\right] \tag{1.1.33.1}$$

1.1.34 Linear Programming Problem (LPP) and Fundamental Corner-Point Theorem. For the canonical problem:

$$\text{Maximize/Minimize: } Z = \mathbf{c}^\top\mathbf{x} \tag{1.1.34.1}$$

$$\text{Subject to: } \mathbf{Ax} \leq \mathbf{b}, \quad \mathbf{x} \geq \mathbf{0} \tag{1.1.34.2}$$

The optimal value of the objective function Z over a bounded convex polygonal feasible region occurs at one of the extreme (corner) vertices of the region.

1.1.35 Law of Total Probability and Bayes' Theorem. If $\{E_1, E_2, \ldots, E_n\}$ forms a partition of sample space S:

$$\Pr(A) = \sum_{j=1}^{n} \Pr(E_j) \Pr(A \mid E_j) \tag{1.1.35.1}$$

$$\Pr(E_k \mid A) = \frac{\Pr(E_k) \Pr(A \mid E_k)}{\sum_{j=1}^{n} \Pr(E_j) \Pr(A \mid E_j)} \tag{1.1.35.2}$$

1.1.36 Distance Between Parallel Planes. For two parallel planes $\Pi_1 : Ax+By+Cz+D_1 = 0$ and $\Pi_2 : Ax + By + Cz + D_2 = 0$:

$$D = \frac{|D_1 - D_2|}{\sqrt{A^2 + B^2 + C^2}} = \frac{|d_1 - d_2|}{\|\mathbf{n}\|} \tag{1.1.36.1}$$

1.1.37 Intersection of a Line and a Plane in 3D. For a line $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$ and plane $\mathbf{r} \cdot \mathbf{n} = d$:

$$(\mathbf{a} + \lambda\mathbf{b}) \cdot \mathbf{n} = d \implies \lambda = \frac{d - \mathbf{a} \cdot \mathbf{n}}{\mathbf{b} \cdot \mathbf{n}}, \quad \mathbf{b} \cdot \mathbf{n} \neq 0 \tag{1.1.37.1}$$

Substituting λ back into the line equation yields the coordinates of the unique point of intersection.

B. Class-12

1.2.1 Find $\left|\mathbf{AB}\right|$, if

$$\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 0 & 2 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 3 & 5 \\ 0 & 0 \end{pmatrix} \tag{1.2.1.1}$$

(12 2019)

Solution: From (1.1.1.1):

$$\left|\mathbf{AB}\right| = \left|\mathbf{A}\right| \left|\mathbf{B}\right| \tag{1.2.1.2}$$

From (1.1.1.2), evaluating the individual determinants:

$$\left|\mathbf{A}\right| = \left| \begin{pmatrix} 0 & -1 \\ 0 & 2 \end{pmatrix} \right| = (0)(2) - (-1)(0) = 0 \tag{1.2.1.3}$$

$$\left|\mathbf{B}\right| = \left| \begin{pmatrix} 3 & 5 \\ 0 & 0 \end{pmatrix} \right| = (3)(0) - (5)(0) = 0 \tag{1.2.1.4}$$

Therefore:

$$\left|\mathbf{AB}\right| = 0 \times 0 = 0 \tag{1.2.1.5}$$

1.2.2 Differentiate e^{3x} with respect to x. (12 2019)

Solution: Let the function be $y = e^{3x}$. From (1.1.2.2), differentiating with respect

to x using the chain rule (1.1.2.1):

$$\frac{dy}{dx} = \frac{d}{dx}\left[e^{3x}\right] = 3e^{3x} \tag{1.2.2.1}$$

1.2.3 Find the order and degree (if defined) of the differential equation

$$\frac{d^2y}{dx^2} + x\left(\frac{dy}{dx}\right)^2 = 2x^2 \ln\left(\frac{d^2y}{dx^2}\right) \tag{1.2.3.1}$$

(12 2019)

Solution: Following the criteria in (1.1.4.1):

- The highest-order derivative appearing in (1.2.3.1) is $\frac{d^2y}{dx^2}$. Hence, Order $= 2$.
- The highest-order derivative $\frac{d^2y}{dx^2}$ appears inside a transcendental logarithmic function $\ln(\cdot)$, meaning the differential equation cannot be expressed as a polynomial in its derivatives. Therefore, the **Degree is not defined**.

1.2.4 Find the direction cosines of a line which makes equal angles with the coordinate axes. (12 2019)

Solution: Let the line make equal angles α with the coordinate axes. From (1.1.4.1) and (1.1.4.2):

$$l^2 + m^2 + n^2 = 1 \implies 3l^2 = 1 \implies l = \pm\frac{1}{\sqrt{3}} \tag{1.2.4.1}$$

From (1.1.4.3), since $l = m = n$, the direction cosines are:

$$\left(\pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}\right) \tag{1.2.4.2}$$

1.2.5 A line passes through the point with position vector $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$ and is in the direction of the vector $\mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}$. Find the equation of the line in Cartesian form. (12 2019)

Solution: From (1.1.5.2), with point $\mathbf{a} = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$ and direction vector $\mathbf{b} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}$:

$$\frac{x-2}{1} = \frac{y-(-1)}{1} = \frac{z-4}{-2} \tag{1.2.5.1}$$

Simplifying:

$$\frac{x-2}{1} = \frac{y+1}{1} = \frac{z-4}{-2} \tag{1.2.5.2}$$

The following code gives the plot

```
codes/12/q5.py
```

1.2.6 Find $\int \left(3 - 2x - x^2\right)\, dx$. (12 2019)

Solution: Let $I = \int \left(3 - 2x - x^2\right)\, dx$. Applying linearity and the power integration rule (1.1.6.1):

$$I = \int 3\, dx - \int 2x\, dx - \int x^2\, dx \tag{1.2.6.1}$$

$$= 3x - 2\left(\frac{x^2}{2}\right) - \frac{x^3}{3} + C \tag{1.2.6.2}$$

$$= 3x - x^2 - \frac{x^3}{3} + C \tag{1.2.6.3}$$

1.2.7 If $\mathbf{A} = \begin{pmatrix} p & 2 \\ 2 & p \end{pmatrix}$ and $\left|\mathbf{A}^3\right| = 125$, find the values of p. (12 2019)

Solution: From (1.1.1.2), evaluating the determinant of $\mathbf{A}$:

$$\left|\mathbf{A}\right| = \left|\begin{pmatrix} p & 2 \\ 2 & p \end{pmatrix}\right| = p^2 - 4 \tag{1.2.7.1}$$

From (1.1.7.1):

$$\left|\mathbf{A}^3\right| = \left(\left|\mathbf{A}\right|\right)^3 = \left(p^2 - 4\right)^3 \tag{1.2.7.2}$$

Given that $\left|\mathbf{A}^3\right| = 125$:

$$\left(p^2 - 4\right)^3 = 5^3 \implies p^2 - 4 = 5 \implies p^2 = 9 \implies p = \pm 3 \tag{1.2.7.3}$$

Thus, the values of p are 3 and -3.

1.2.8 Examine whether the operation $*$ defined on the set of all real numbers by $a * b = a^2 + b^2$ is a binary operation or not, and if it is a binary operation, find whether it is associative or not. (12 2019)

Solution: From (1.1.8.1), for any $a, b \in \mathbb{R}$, $a^2 \in \mathbb{R}$ and $b^2 \in \mathbb{R}$. Since the sum of real numbers is real, $a^2 + b^2 \in \mathbb{R}$. Hence, $*$ is a well-defined binary operation on $\mathbb{R}$. To test associativity from (1.1.8.2):

$$\text{LHS:} \quad (a * b) * c = \left(a^2 + b^2\right) * c = \left(a^2 + b^2\right)^2 + c^2 \tag{1.2.8.1}$$

$$\text{RHS:} \quad a * (b * c) = a * \left(b^2 + c^2\right) = a^2 + \left(b^2 + c^2\right)^2 \tag{1.2.8.2}$$

Choosing counterexample $a=1, b=2, c=3 \in \mathbb{R}$:

$$(1*2)*3 = \left(1^2+2^2\right)*3 = 5*3 = 5^2+3^2 = 25+9 = 34 \tag{1.2.8.3}$$

$$1*(2*3) = 1*\left(2^2+3^2\right) = 1*13 = 1^2+13^2 = 1+169 = 170 \tag{1.2.8.4}$$

Since $34 \neq 170$, the operation $*$ is **not associative**.

1.2.9 If $\|\mathbf{a}\| = 2$, $\|\mathbf{b}\| = 7$, and $\mathbf{a}\times\mathbf{b} = \begin{pmatrix}3\\2\\6\end{pmatrix}$, find the angle between $\mathbf{a}$ and $\mathbf{b}$. (12 2019)

Solution: Let θ be the angle between $\mathbf{a}$ and $\mathbf{b}$. Evaluating the magnitude of $\mathbf{a}\times\mathbf{b}$:

$$\|\mathbf{a}\times\mathbf{b}\| = \sqrt{3^2+2^2+6^2} = \sqrt{9+4+36} = \sqrt{49} = 7 \tag{1.2.9.1}$$

From (1.1.9.1) and (1.1.9.2):

$$\sin\theta = \frac{\|\mathbf{a}\times\mathbf{b}\|}{\|\mathbf{a}\|\,\|\mathbf{b}\|} = \frac{7}{2\times 7} = \frac{1}{2} \tag{1.2.9.2}$$

Since $\theta \in [0,\pi]$:

$$\theta = \frac{\pi}{6} \quad \text{or} \quad \theta = \frac{5\pi}{6} \tag{1.2.9.3}$$

1.2.10 Find the volume of a cuboid whose edges are given by $\mathbf{a} = \begin{pmatrix}-3\\7\\5\end{pmatrix}$, $\mathbf{b} = \begin{pmatrix}-5\\7\\-3\end{pmatrix}$, and $\mathbf{c} = \begin{pmatrix}7\\-5\\-3\end{pmatrix}$. (12 2019)

Solution: From (1.1.10.1), the volume V of a parallelepiped (cuboid) is the absolute value of the scalar triple product:

$$V = \left|\begin{matrix}\mathbf{a} & \mathbf{b} & \mathbf{c}\end{matrix}\right| = \left|\begin{vmatrix}-3 & -5 & 7\\ 7 & 7 & -5\\ 5 & -3 & -3\end{vmatrix}\right| \tag{1.2.10.1}$$

Expanding the determinant:

$$\begin{vmatrix}\mathbf{a} & \mathbf{b} & \mathbf{c}\end{vmatrix} = -3\begin{vmatrix}7 & -5\\ -3 & -3\end{vmatrix} - (-5)\begin{vmatrix}7 & -5\\ 5 & -3\end{vmatrix} + 7\begin{vmatrix}7 & 7\\ 5 & -3\end{vmatrix} \tag{1.2.10.2}$$

$$= -3(-21-15) + 5(-21-(-25)) + 7(-21-35) \tag{1.2.10.3}$$

$$= -3(-36) + 5(4) + 7(-56) = 108+20-392 = -264 \tag{1.2.10.4}$$

Taking the absolute value:

$$V = |-264| = 264 \text{ cubic units} \tag{1.2.10.5}$$

The following code verifies the volume:

```
codes/12/q10.py
```

1.2.11 A coin is tossed 5 times. What is the probability of getting (i) 3 heads, (ii) at most

3 heads? (12 2019)

Solution: Let X denote the number of heads obtained in $n = 5$ independent coin tosses. This represents a sequence of Bernoulli trials with success probability $p = \frac{1}{2}$ and failure probability $q = 1 - p = \frac{1}{2}$. From the binomial distribution formula (1.1.11.1):

$$\Pr(X = k) = \binom{n}{k} p^k q^{n-k} = \binom{5}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{5-k} = \frac{\binom{5}{k}}{32} \tag{1.2.11.1}$$

(i) Probability of getting exactly 3 heads ($k = 3$):

$$\Pr(X = 3) = \frac{\binom{5}{3}}{32} = \frac{10}{32} = \frac{5}{16} \tag{1.2.11.2}$$

(ii) Probability of getting at most 3 heads ($\Pr(X \leq 3)$): Using the complement relation from (1.1.11.2):

$$\Pr(X \leq 3) = 1 - [\Pr(X = 4) + \Pr(X = 5)] \tag{1.2.11.3}$$

$$= 1 - \left(\frac{\binom{5}{4}}{32} + \frac{\binom{5}{5}}{32}\right) = 1 - \frac{5+1}{32} = 1 - \frac{6}{32} = \frac{26}{32} = \frac{13}{16} \tag{1.2.11.4}$$

1.2.12 Find the probability distribution of X, the number of heads in a simultaneous toss of two coins. (12 2019)

Solution: When two fair coins are tossed simultaneously, the sample space S is:

$$S = \{\text{HH, HT, TH, TT}\} \tag{1.2.12.1}$$

Let X be the random variable representing the number of heads. The possible values of X are $0, 1, 2$. From (1.1.12.1):

$$\Pr(X = 0) = \frac{n(\{\text{TT}\})}{n(S)} = \frac{1}{4} \tag{1.2.12.2}$$

$$\Pr(X = 1) = \frac{n(\{\text{HT, TH}\})}{n(S)} = \frac{2}{4} = \frac{1}{2} \tag{1.2.12.3}$$

$$\Pr(X = 2) = \frac{n(\{\text{HH}\})}{n(S)} = \frac{1}{4} \tag{1.2.12.4}$$

Verifying the distribution sum:

$$\sum_{x=0}^{2} \Pr(X = x) = \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1 \tag{1.2.12.5}$$

1.2.13 Find $\displaystyle\int \frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x}\, dx$. (12 2019)

Solution: Let $I = \int \frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x}\, dx$. Splitting the fraction into two terms:

$$\frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x} = \frac{\sin^3 x}{\sin^2 x \cos^2 x} + \frac{\cos^3 x}{\sin^2 x \cos^2 x} \tag{1.2.13.1}$$

$$= \frac{\sin x}{\cos^2 x} + \frac{\cos x}{\sin^2 x} \tag{1.2.13.2}$$

$$= \left(\frac{\sin x}{\cos x}\right)\left(\frac{1}{\cos x}\right) + \left(\frac{\cos x}{\sin x}\right)\left(\frac{1}{\sin x}\right) \tag{1.2.13.3}$$

$$= \sec x \tan x + \csc x \cot x \tag{1.2.13.4}$$

Integrating term-by-term using (1.1.13.1) and (1.1.13.2):

$$I = \int (\sec x \tan x + \csc x \cot x)\, dx \tag{1.2.13.5}$$

$$= \int \sec x \tan x\, dx + \int \csc x \cot x\, dx \tag{1.2.13.6}$$

$$= \sec x - \csc x + C \tag{1.2.13.7}$$

1.2.14 Find $\displaystyle\int \frac{x-3}{(x-1)^3} e^x\, dx$. (12 2019)

Solution: Let $I = \int \frac{x-3}{(x-1)^3} e^x\, dx$. Expressing the numerator in terms of $(x-1)$:

$$\frac{x-3}{(x-1)^3} = \frac{(x-1)-2}{(x-1)^3} = \frac{1}{(x-1)^2} - \frac{2}{(x-1)^3} \tag{1.2.14.1}$$

Let $f(x) = \frac{1}{(x-1)^2} = (x-1)^{-2}$. Its derivative is:

$$f'(x) = -2(x-1)^{-3} = -\frac{2}{(x-1)^3} \tag{1.2.14.2}$$

Using the standard exponential integral formula (1.1.14.1):

$$\int e^x [f(x) + f'(x)]\, dx = e^x f(x) + C \tag{1.2.14.3}$$

Substituting $f(x)$:

$$I = \int e^x \left[\frac{1}{(x-1)^2} - \frac{2}{(x-1)^3}\right] dx = \frac{e^x}{(x-1)^2} + C \tag{1.2.14.4}$$

1.2.15 If $\Pr(\text{not } A) = 0.7$, $\Pr(B) = 0.7$, and $\Pr(B \mid A) = 0.5$, find $\Pr(A \mid B)$. (12 2019)

Solution: From the complement rule, $\Pr(A) = 1 - \Pr(\text{not } A) = 1 - 0.7 = 0.3$. From the definition of conditional probability (1.1.15.1):

$$\Pr(B \mid A) = \frac{\Pr(A \cap B)}{\Pr(A)} \implies 0.5 = \frac{\Pr(A \cap B)}{0.3} \implies \Pr(A \cap B) = 0.15 \tag{1.2.15.1}$$

Now, computing $\Pr(A \mid B)$ via (1.1.15.1):

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)} = \frac{0.15}{0.7} = \frac{15}{70} = \frac{3}{14} \approx 0.2143 \tag{1.2.15.2}$$

1.2.16 Find the general solution of the differential equation $\dfrac{dy}{dx} = e^{x+y}$. (12 2019)

Solution: Using index laws, the differential equation can be separated:

$$\frac{dy}{dx} = e^x e^y \implies e^{-y}\,dy = e^x\,dx \tag{1.2.16.1}$$

Integrating both sides using (1.1.16.1):

$$\int e^{-y}\,dy = \int e^x\,dx \implies -e^{-y} = e^x + C_1 \tag{1.2.16.2}$$

Rearranging terms:

$$e^x + e^{-y} = C \quad \text{where } C = -C_1 \tag{1.2.16.3}$$

1.2.17 Find the value of $\sin\left(\cos^{-1}\frac{4}{5} + \tan^{-1}\frac{2}{3}\right)$. (12 2019)

Solution: From (1.1.17.1), converting $\cos^{-1}\left(\frac{4}{5}\right)$ to $\tan^{-1}$:

$$\cos^{-1}\frac{4}{5} = \tan^{-1}\left(\frac{\sqrt{1-(4/5)^2}}{4/5}\right) = \tan^{-1}\left(\frac{3/5}{4/5}\right) = \tan^{-1}\frac{3}{4} \tag{1.2.17.1}$$

Using the addition formula (1.1.17.2) for $\tan^{-1}\frac{3}{4} + \tan^{-1}\frac{2}{3}$ (since $uv = \frac{3}{4}\cdot\frac{2}{3} = \frac{1}{2} < 1$):

$$\tan^{-1}\frac{3}{4} + \tan^{-1}\frac{2}{3} = \tan^{-1}\left(\frac{\frac{3}{4}+\frac{2}{3}}{1-\frac{3}{4}\cdot\frac{2}{3}}\right) \tag{1.2.17.2}$$

$$= \tan^{-1}\left(\frac{\frac{17}{12}}{1-\frac{6}{12}}\right) = \tan^{-1}\left(\frac{17/12}{6/12}\right) = \tan^{-1}\frac{17}{6} \tag{1.2.17.3}$$

Let $\theta = \tan^{-1}\frac{17}{6}$. From (1.1.17.3):

$$\sin\left(\tan^{-1}\frac{17}{6}\right) = \frac{17}{\sqrt{17^2+6^2}} = \frac{17}{\sqrt{289+36}} = \frac{17}{\sqrt{325}} = \frac{17}{5\sqrt{13}} = \frac{17\sqrt{13}}{65} \tag{1.2.17.4}$$

1.2.18 Using properties of determinants, show that

$$\begin{vmatrix} 3a & -a+b & -a+c \\ -b+a & 3b & -b+c \\ -c+a & -c+b & 3c \end{vmatrix} = 3(a+b+c)(ab+bc+ca) \tag{1.2.18.1}$$

(12 2019)

Solution: Let Δ denote the given determinant. Applying the column operation $C_1 \to$

$C_1 + C_2 + C_3$ in accordance with (1.1.18.1):

$$\Delta = \begin{vmatrix} a+b+c & -a+b & -a+c \\ a+b+c & 3b & -b+c \\ a+b+c & -c+b & 3c \end{vmatrix} \tag{1.2.18.2}$$

Factoring out $(a+b+c)$ from column C_1:

$$\Delta = (a+b+c) \begin{vmatrix} 1 & -a+b & -a+c \\ 1 & 3b & -b+c \\ 1 & -c+b & 3c \end{vmatrix} \tag{1.2.18.3}$$

Applying row operations $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$:

$$\Delta = (a+b+c) \begin{vmatrix} 1 & -a+b & -a+c \\ 0 & a+2b & a-b \\ 0 & a-c & a+2c \end{vmatrix} \tag{1.2.18.4}$$

Expanding along column C_1:

$$\Delta = (a+b+c) \begin{vmatrix} a+2b & a-b \\ a-c & a+2c \end{vmatrix} \tag{1.2.18.5}$$

$$= (a+b+c)\left[(a+2b)(a+2c) - (a-b)(a-c)\right] \tag{1.2.18.6}$$

$$= (a+b+c)\left[\left(a^2+2ac+2ab+4bc\right) - \left(a^2-ac-ab+bc\right)\right] \tag{1.2.18.7}$$

$$= (a+b+c)(3ab+3bc+3ca) \tag{1.2.18.8}$$

$$= 3(a+b+c)(ab+bc+ca) \tag{1.2.18.9}$$

1.2.19 Check whether the relation R defined on the set $A = \{1,2,3,4,5,6\}$ by $R = \{(a,b) : b = a+1\}$ is reflexive, symmetric or transitive. (12 2019)

Solution:

Listing all ordered pairs satisfying $b = a+1$ for $a, b \in A$:

$$R = \{(1,2), (2,3), (3,4), (4,5), (5,6)\} \tag{1.2.19.1}$$

- **Reflexivity:** A relation is reflexive if $(a,a) \in R$ for all $a \in A$. Since $(1,1) \notin R$, the relation is **not reflexive**.
- **Symmetry:** A relation is symmetric if $(a,b) \in R \implies (b,a) \in R$. Since $(1,2) \in R$ but $(2,1) \notin R$, the relation is **not symmetric**.
- **Transitivity:** A relation is transitive if $(a,b) \in R$ and $(b,c) \in R \implies (a,c) \in R$. Since $(1,2) \in R$ and $(2,3) \in R$ but $(1,3) \notin R$, the relation is **not transitive**.

Thus, R is neither reflexive, nor symmetric, nor transitive.

1.2.20 Let $f : \mathbb{N} \to Y$ be a function defined as $f(x) = 4x+3$, where $Y = \{y \in \mathbb{N} : y = 4x+3 \text{ for some } x \in \mathbb{N}\}$. Show that f is invertible. Find its inverse. (12 2019)

Solution: From (1.1.20.1), to prove f is injective (one-one), let $f(x_1) = f(x_2)$ for $x_1, x_2 \in \mathbb{N}$:

$$4x_1 + 3 = 4x_2 + 3 \implies 4x_1 = 4x_2 \implies x_1 = x_2 \tag{1.2.20.1}$$

Thus, f is injective. From (1.1.20.2), for any $y \in Y$, by definition $y = 4x+3$ for

some $x \in \mathbb{N}$. Solving for x:

$$x = \frac{y-3}{4} \in \mathbb{N} \tag{1.2.20.2}$$

Since every $y \in Y$ has a pre-image in $\mathbb{N}$, f is surjective (onto). Since f is both injective and surjective, it is bijective and hence invertible. Defining the inverse function $f^{-1} : Y \to \mathbb{N}$:

$$f^{-1}(y) = \frac{y-3}{4} \tag{1.2.20.3}$$

1.2.21 If $x\sqrt{1+y} + y\sqrt{1+x} = 0$, where $x \neq y$, prove that $\dfrac{dy}{dx} = -\dfrac{1}{(x+1)^2}$. (12 2019)

Solution: Rearranging the given equation:

$$x\sqrt{1+y} = -y\sqrt{1+x} \tag{1.2.21.1}$$

Squaring both sides:

$$x^2(1+y) = y^2(1+x) \tag{1.2.21.2}$$
$$x^2 + x^2y = y^2 + xy^2 \tag{1.2.21.3}$$
$$x^2 - y^2 + x^2y - xy^2 = 0 \tag{1.2.21.4}$$
$$(x-y)(x+y) + xy(x-y) = 0 \tag{1.2.21.5}$$
$$(x-y)[x+y+xy] = 0 \tag{1.2.21.6}$$

Since $x \neq y$, we have $x - y \neq 0$, which implies:

$$x + y + xy = 0 \implies y(1+x) = -x \implies y = -\frac{x}{1+x} \tag{1.2.21.7}$$

Rewriting y to facilitate differentiation:

$$y = -\frac{(x+1)-1}{x+1} = -\left(1 - \frac{1}{x+1}\right) = -1 + (x+1)^{-1} \tag{1.2.21.8}$$

Differentiating with respect to x:

$$\frac{dy}{dx} = \frac{d}{dx}\left[-1 + (x+1)^{-1}\right] = -(x+1)^{-2} \cdot 1 = -\frac{1}{(x+1)^2} \tag{1.2.21.9}$$

1.2.22 If $(\cos x)^y = (\sin y)^x$, find $\dfrac{dy}{dx}$. (12 2019)

Solution: Taking the natural logarithm on both sides using (1.1.21.1):

$$\ln\left((\cos x)^y\right) = \ln\left((\sin y)^x\right) \tag{1.2.22.1}$$
$$y \ln(\cos x) = x \ln(\sin y) \tag{1.2.22.2}$$

Differentiating both sides implicitly with respect to x using (1.1.21.2):

$$\frac{dy}{dx}\ln(\cos x)+y\cdot\frac{-\sin x}{\cos x}=1\cdot\ln(\sin y)+x\cdot\frac{\cos y}{\sin y}\frac{dy}{dx} \tag{1.2.22.3}$$

$$\frac{dy}{dx}\ln(\cos x)-y\tan x=\ln(\sin y)+x\cot y\frac{dy}{dx} \tag{1.2.22.4}$$

Grouping the derivative terms:

$$\frac{dy}{dx}\left[\ln(\cos x)-x\cot y\right]=\ln(\sin y)+y\tan x \tag{1.2.22.5}$$

Solving for $\dfrac{dy}{dx}$:

$$\frac{dy}{dx}=\frac{\ln(\sin y)+y\tan x}{\ln(\cos x)-x\cot y} \tag{1.2.22.6}$$

1.2.23 Prove that $\displaystyle\int_0^a f(x)\,dx=\int_0^a f(a-x)\,dx$, and hence evaluate $\displaystyle\int_0^{\pi/2}\frac{x}{\sin x+\cos x}\,dx$. (12 2019)

Solution: To prove King's property (1.1.22.1), let $t=a-x\implies dx=-dt$. When $x=0$, $t=a$; when $x=a$, $t=0$:

$$\int_0^a f(x)\,dx=\int_a^0 f(a-t)\,(-dt)=\int_0^a f(a-t)\,dt=\int_0^a f(a-x)\,dx \tag{1.2.23.1}$$

Now, evaluating the given definite integral:

$$I=\int_0^{\pi/2}\frac{x}{\sin x+\cos x}\,dx \tag{1.2.23.2}$$

Applying property (1.1.22.1):

$$I=\int_0^{\pi/2}\frac{\frac{\pi}{2}-x}{\sin\left(\frac{\pi}{2}-x\right)+\cos\left(\frac{\pi}{2}-x\right)}\,dx \tag{1.2.23.3}$$

$$=\int_0^{\pi/2}\frac{\frac{\pi}{2}-x}{\cos x+\sin x}\,dx \tag{1.2.23.4}$$

Adding (1.2.23.2) and (1.2.23.4):

$$2I=\int_0^{\pi/2}\frac{x+\frac{\pi}{2}-x}{\sin x+\cos x}\,dx=\frac{\pi}{2}\int_0^{\pi/2}\frac{1}{\sin x+\cos x}\,dx \tag{1.2.23.5}$$

$$I=\frac{\pi}{4}\int_0^{\pi/2}\frac{1}{\sqrt{2}\cos\left(x-\frac{\pi}{4}\right)}\,dx=\frac{\pi}{4\sqrt{2}}\int_0^{\pi/2}\sec\left(x-\frac{\pi}{4}\right)\,dx \tag{1.2.23.6}$$

Substituting $u = x - \frac{\pi}{4}$:

$$I = \frac{\pi}{4\sqrt{2}} \int_{-\pi/4}^{\pi/4} \sec u \, du = \frac{\pi}{4\sqrt{2}} \cdot 2 \ln \left| \sec \frac{\pi}{4} + \tan \frac{\pi}{4} \right| \tag{1.2.23.7}$$

$$= \frac{\pi}{2\sqrt{2}} \ln \left(\sqrt{2} + 1 \right) \tag{1.2.23.8}$$

1.2.24 Find $\displaystyle\int \frac{x^2 + x + 1}{(x+2)(x^2+1)} \, dx$. (12 2019)

Solution: Using partial fraction decomposition from (1.1.23.1):

$$\frac{x^2 + x + 1}{(x+2)(x^2+1)} = \frac{A}{x+2} + \frac{Bx + C}{x^2+1} \tag{1.2.24.1}$$

Multiplying by $(x+2)(x^2+1)$:

$$x^2 + x + 1 = A(x^2+1) + (Bx + C)(x+2) \tag{1.2.24.2}$$

$$= (A+B)x^2 + (2B+C)x + (A+2C) \tag{1.2.24.3}$$

Comparing coefficients of like powers of x:

$$A + B = 1 \tag{1.2.24.4}$$

$$2B + C = 1 \tag{1.2.24.5}$$

$$A + 2C = 1 \tag{1.2.24.6}$$

Solving this linear system yields $A = \frac{3}{5}$, $B = \frac{2}{5}$, and $C = \frac{1}{5}$. Thus:

$$\frac{x^2 + x + 1}{(x+2)(x^2+1)} = \frac{3}{5}\left(\frac{1}{x+2}\right) + \frac{1}{5}\left(\frac{2x}{x^2+1}\right) + \frac{1}{5}\left(\frac{1}{x^2+1}\right) \tag{1.2.24.7}$$

Integrating term-by-term using (1.1.23.2)–(1.1.23.4):

$$I = \frac{3}{5}\int \frac{1}{x+2} \, dx + \frac{1}{5}\int \frac{2x}{x^2+1} \, dx + \frac{1}{5}\int \frac{1}{x^2+1} \, dx \tag{1.2.24.8}$$

$$= \frac{3}{5}\ln|x+2| + \frac{1}{5}\ln(x^2+1) + \frac{1}{5}\tan^{-1} x + C \tag{1.2.24.9}$$

1.2.25 Solve the differential equation $x\dfrac{dy}{dx} = y - x\tan\left(\frac{y}{x}\right)$. (12 2019)

Solution: Rewriting in the homogeneous form (1.1.24.1):

$$\frac{dy}{dx} = \frac{y}{x} - \tan\left(\frac{y}{x}\right) \tag{1.2.25.1}$$

Substituting $y = vx$ and $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$ in accordance with (1.1.24.2):

$$v + x\frac{dv}{dx} = v - \tan v \implies x\frac{dv}{dx} = -\tan v \tag{1.2.25.2}$$

Separating variables:

$$\frac{dv}{\tan v} = -\frac{dx}{x} \implies \cot v\, dv = -\frac{dx}{x} \tag{1.2.25.3}$$

Integrating both sides:

$$\ln|\sin v| = -\ln|x| + \ln C_1 \implies \ln|x \sin v| = \ln C_1 \implies x \sin v = C \tag{1.2.25.4}$$

Substituting back $v = \frac{y}{x}$:

$$x \sin\left(\frac{y}{x}\right) = C \tag{1.2.25.5}$$

1.2.26 Solve the differential equation $\dfrac{dy}{dx} = -\dfrac{x + y\cos x}{1 + \sin x}$. (12 2019)

Solution: Rearranging into the standard linear differential form (1.1.25.1):

$$\frac{dy}{dx} + \left(\frac{\cos x}{1+\sin x}\right) y = -\frac{x}{1+\sin x} \tag{1.2.26.1}$$

Here $P(x) = \frac{\cos x}{1+\sin x}$ and $Q(x) = -\frac{x}{1+\sin x}$. Finding the integrating factor via (1.1.25.3):

$$\text{I.F.} = e^{\int \frac{\cos x}{1+\sin x}\, dx} = e^{\ln(1+\sin x)} = 1 + \sin x \tag{1.2.26.2}$$

Multiplying through and integrating:

$$y\,(1+\sin x) = \int \left(-\frac{x}{1+\sin x}\right)(1+\sin x)\, dx + C \tag{1.2.26.3}$$

$$= \int (-x)\, dx + C = -\frac{x^2}{2} + C \tag{1.2.26.4}$$

Multiplying by 2:

$$2y\,(1+\sin x) + x^2 = C' \tag{1.2.26.5}$$

1.2.27 The scalar product of the vector $\mathbf{a} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ with a unit vector along the sum of the vectors $\mathbf{b} = 2\mathbf{i} + 4\mathbf{j} - 5\mathbf{k}$ and $\mathbf{c} = \lambda\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ is equal to 1. Find the value of λ and hence find the unit vector along $\mathbf{b} + \mathbf{c}$. (12 2019)

Solution: Summing vectors $\mathbf{b}$ and $\mathbf{c}$:

$$\mathbf{b} + \mathbf{c} = \begin{pmatrix} 2+\lambda \\ 6 \\ -2 \end{pmatrix} \tag{1.2.27.1}$$

The unit vector $\mathbf{u}$ along $\mathbf{b} + \mathbf{c}$ is given by (1.1.26.1):

$$\mathbf{u} = \frac{\mathbf{b}+\mathbf{c}}{\|\mathbf{b}+\mathbf{c}\|} = \frac{\begin{pmatrix} \lambda+2 \\ 6 \\ -2 \end{pmatrix}}{\sqrt{(\lambda+2)^2 + 36 + 4}} = \frac{\begin{pmatrix} \lambda+2 \\ 6 \\ -2 \end{pmatrix}}{\sqrt{(\lambda+2)^2 + 40}} \tag{1.2.27.2}$$

Given that $\mathbf{a}\cdot\mathbf{u}=1$:

$$\begin{pmatrix}1 & 1 & 1\end{pmatrix}\cdot\frac{\begin{pmatrix}\lambda+2\\6\\-2\end{pmatrix}}{\sqrt{(\lambda+2)^2+40}}=1\implies\frac{(\lambda+2)+6-2}{\sqrt{(\lambda+2)^2+40}}=1\implies\frac{\lambda+6}{\sqrt{(\lambda+2)^2+40}}=1 \tag{1.2.27.3}$$

Squaring both sides ($\lambda+6>0$):

$$(\lambda+6)^2=(\lambda+2)^2+40\implies\lambda^2+12\lambda+36=\lambda^2+4\lambda+4+40\implies 8\lambda=8\implies\lambda \tag{1.2.27.4}$$

Substituting $\lambda=1$:

$$\mathbf{b}+\mathbf{c}=\begin{pmatrix}3\\6\\-2\end{pmatrix},\quad\|\mathbf{b}+\mathbf{c}\|=\sqrt{9+36+4}=\sqrt{49}=7 \tag{1.2.27.5}$$

Thus, the unit vector along $\mathbf{b}+\mathbf{c}$ is:

$$\mathbf{u}=\frac{1}{7}\begin{pmatrix}3\\6\\-2\end{pmatrix}=\frac{3}{7}\mathbf{i}+\frac{6}{7}\mathbf{j}-\frac{2}{7}\mathbf{k} \tag{1.2.27.6}$$

1.2.28 If $(a+bx)\,e^{y/x}=x$, then prove that $x^3\dfrac{d^2y}{dx^2}=\left(x\dfrac{dy}{dx}-y\right)^2$. (12 2019)

Solution: Taking the natural logarithm on both sides:

$$\ln(a+bx)+\frac{y}{x}=\ln x\implies\frac{y}{x}=\ln x-\ln(a+bx) \tag{1.2.28.1}$$

Differentiating with respect to x:

$$\frac{x\frac{dy}{dx}-y}{x^2}=\frac{1}{x}-\frac{b}{a+bx}=\frac{a+bx-bx}{x(a+bx)}=\frac{a}{x(a+bx)} \tag{1.2.28.2}$$

Multiplying by x^2:

$$x\frac{dy}{dx}-y=\frac{ax}{a+bx} \tag{1.2.28.3}$$

Differentiating (1.2.28.3) with respect to x:

$$\frac{dy}{dx}+x\frac{d^2y}{dx^2}-\frac{dy}{dx}=\frac{(a+bx)(a)-(ax)(b)}{(a+bx)^2} \tag{1.2.28.4}$$

$$x\frac{d^2y}{dx^2}=\frac{a^2}{(a+bx)^2} \tag{1.2.28.5}$$

Multiplying both sides by x^2:

$$x^3\frac{d^2y}{dx^2} = \frac{a^2x^2}{(a+bx)^2} = \left(\frac{ax}{a+bx}\right)^2 \tag{1.2.28.6}$$

Substituting (1.2.28.3):

$$x^3\frac{d^2y}{dx^2} = \left(x\frac{dy}{dx} - y\right)^2 \tag{1.2.28.7}$$

The following code verifies the differential identity:

codes/12/q28.py

1.2.29 The volume of a cube is increasing at the rate of 8 cm^3/s. How fast is the surface area increasing when the length of its edge is 12 cm? (12 2019)

Solution: Let x be the edge length of the cube. From (1.1.27.2):

$$V = x^3 \implies \frac{dV}{dt} = 3x^2\frac{dx}{dt} \tag{1.2.29.1}$$

Given $\frac{dV}{dt} = 8$ cm^3/s:

$$8 = 3x^2\frac{dx}{dt} \implies \frac{dx}{dt} = \frac{8}{3x^2} \tag{1.2.29.2}$$

Surface area $S = 6x^2$. From (1.1.27.2), the rate of change of surface area is:

$$\frac{dS}{dt} = 12x\frac{dx}{dt} = 12x\left(\frac{8}{3x^2}\right) = \frac{32}{x} \tag{1.2.29.3}$$

Evaluating at $x = 12$ cm:

$$\left.\frac{dS}{dt}\right|_{x=12} = \frac{32}{12} = \frac{8}{3}\ \text{cm}^2/\text{s} \tag{1.2.29.4}$$

1.2.30 Find the Cartesian and vector equations of the plane passing through the points $A(2,5,-3)$, $B(-2,-3,5)$, and $C(5,3,-3)$. (12 2019)

Solution: Let the position vectors of the points be

$$\mathbf{a} = \begin{pmatrix}2\\5\\-3\end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix}-2\\-3\\5\end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix}5\\3\\-3\end{pmatrix} \tag{1.2.30.1}$$

Evaluating displacement vectors lying in the plane:

$$\mathbf{AB} = \mathbf{b} - \mathbf{a} = \begin{pmatrix}-4\\-8\\8\end{pmatrix} \tag{1.2.30.2}$$

$$\mathbf{AC} = \mathbf{c} - \mathbf{a} = \begin{pmatrix}3\\-2\\0\end{pmatrix} \tag{1.2.30.3}$$

From (1.1.28.2), the normal vector $\mathbf{n}$ is obtained by cross product:

$$\mathbf{n} = \mathbf{AB} \times \mathbf{AC} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -4 & -8 & 8 \\ 3 & -2 & 0 \end{vmatrix} = \mathbf{i}\left(0-(-16)\right) - \mathbf{j}\left(0-24\right) + \mathbf{k}\left(8-(-24)\right) = 16\mathbf{i} + 2 \tag{1.2.30.4}$$

Scaling by $\frac{1}{8}$ yields the simplified normal vector:

$$\mathbf{n}_0 = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} \tag{1.2.30.5}$$

From (1.1.28.2), evaluating $\mathbf{a} \cdot \mathbf{n}_0$:

$$\begin{pmatrix} 2 & 5 & -3 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} = 4 + 15 - 12 = 7 \tag{1.2.30.6}$$

Thus, the vector equation of the plane is:

$$\mathbf{r} \cdot \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} = 7 \tag{1.2.30.7}$$

Substituting $\mathbf{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, the Cartesian equation is:

$$2x + 3y + 4z = 7 \tag{1.2.30.8}$$

The following code verifies the plane equations:

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codes/12/q30.py
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1.2.31 Find the point on the curve $y^2 = 4x$ which is nearest to the point $(2, -8)$. (12 2019)

Solution: Let $P\left(x, y\right)$ be a point on the parabola $y^2 = 4x$, so its coordinates can be expressed parametrically as $\begin{pmatrix} y^2/4 \\ y \end{pmatrix}$. Let the fixed point be $\mathbf{A} = \begin{pmatrix} 2 \\ -8 \end{pmatrix}$. The square of the Euclidean distance Z between P and A from (1.1.29.1) is:

$$Z = \left(\frac{y^2}{4} - 2\right)^2 + \left(y - (-8)\right)^2 = \left(\frac{y^2}{4} - 2\right)^2 + \left(y + 8\right)^2 \tag{1.2.31.1}$$

Differentiating Z with respect to y:

$$\frac{dZ}{dy} = 2\left(\frac{y^2}{4} - 2\right)\left(\frac{2y}{4}\right) + 2\left(y + 8\right) \tag{1.2.31.2}$$

$$= \frac{y^3}{4} - 2y + 2y + 16 = \frac{y^3}{4} + 16 \tag{1.2.31.3}$$

Setting $\frac{dZ}{dy} = 0$ for stationary points:

$$\frac{y^3}{4} + 16 = 0 \implies y^3 = -64 \implies y = -4 \tag{1.2.31.4}$$

Applying the second derivative test (1.1.29.2):

$$\frac{d^2Z}{dy^2} = \frac{3y^2}{4} \tag{1.2.31.5}$$

Evaluating at $y = -4$:

$$\left.\frac{d^2Z}{dy^2}\right|_{y=-4} = \frac{3\left(-4\right)^2}{4} = \frac{3\left(16\right)}{4} = 12 > 0 \tag{1.2.31.6}$$

Since $\frac{d^2Z}{dy^2} > 0$, Z attains its minimum at $y = -4$. Substituting $y = -4$ into $x = \frac{y^2}{4}$:

$$x = \frac{\left(-4\right)^2}{4} = \frac{16}{4} = 4 \tag{1.2.31.7}$$

Therefore, the point on the curve nearest to $(2, -8)$ is $(4, -4)$. The following code verifies the optimization:

codes/12/q31.py

1.2.32 Find $\int_1^3 \left(x^2 + 2 + e^{2x}\right)\,dx$ as the limit of sums. (12 2019)

Solution: From the Riemann sum definition (1.1.30.1):

$$\int_a^b f(x)\,dx = \lim_{h\to 0} h\sum_{r=1}^{n} f\left(a + rh\right) \tag{1.2.32.1}$$

where $a = 1, b = 3 \implies nh = b - a = 2$, and $f(x) = x^2 + 2 + e^{2x}$. Evaluating $f\left(1 + rh\right)$:

$$f\left(1 + rh\right) = \left(1 + rh\right)^2 + 2 + e^{2(1+rh)} = 3 + 2rh + r^2h^2 + e^2\left(e^{2h}\right)^r \tag{1.2.32.2}$$

Using standard summation formulas (1.1.30.2), (1.1.30.3), and (1.1.30.4):

$$hS_n = 3\left(nh\right) + 2h^2\left[\frac{n\left(n+1\right)}{2}\right] + h^3\left[\frac{n\left(n+1\right)\left(2n+1\right)}{6}\right] + e^2e^{2h}\left(\frac{e^{2nh}-1}{e^{2h}-1}\right)h \tag{1.2.32.3}$$

$$= 3\left(nh\right) + \left(nh\right)\left(nh+h\right) + \frac{\left(nh\right)\left(nh+h\right)\left(2nh+h\right)}{6} + e^2\left(e^{2nh}-1\right)e^{2h}\left(\frac{2h}{e^{2h}-1}\right) \tag{1.2.32.4}$$

Taking the limit as $h \to 0$ using (1.1.30.5):

$$\int_1^3 \left(x^2 + 2 + e^{2x}\right) dx = 3\,(2) + 2\,(2) + \frac{2\,(2)\,(4)}{6} + \left(e^6 - e^2\right)(1)\left(\frac{1}{2} \cdot 1\right) \tag{1.2.32.5}$$

$$= 6 + 4 + \frac{8}{3} + \frac{e^6 - e^2}{2} = \frac{38}{3} + \frac{e^6 - e^2}{2} \tag{1.2.32.6}$$

1.2.33 Using integration, find the area of the triangular region whose sides have the equations $y = 2x + 1$, $y = 3x + 1$, and $x = 4$. (12 2019)

Solution: Solving the lines simultaneously:

- Intersection of $y = 2x + 1$ and $y = 3x + 1 \implies x = 0, y = 1$ (Vertex $A(0, 1)$).
- Intersection of $y = 2x + 1$ and $x = 4 \implies x = 4, y = 9$ (Vertex $B(4, 9)$).
- Intersection of $y = 3x + 1$ and $x = 4 \implies x = 4, y = 13$ (Vertex $C(4, 13)$).

From (1.1.31.1), integrating between $x = 0$ and $x = 4$:

$$\text{Area} = \int_0^4 \left[(3x + 1) - (2x + 1)\right] dx = \int_0^4 x\, dx \tag{1.2.33.1}$$

$$= \frac{x^2}{2}\bigg|_0^4 = \frac{4^2}{2} = 8 \text{ square units} \tag{1.2.33.2}$$

The following code verifies the area:

codes/12/q33.py

1.2.34 If $\mathbf{A} = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 1 & 2 \\ 5 & 1 & 1 \end{bmatrix}$, find $\mathbf{A}^{-1}$. Hence solve the system of equations:

$$x + 3y + 4z = 8 \tag{1.2.34.1}$$

$$2x + y + 2z = 5 \tag{1.2.34.2}$$

$$5x + y + z = 7 \tag{1.2.34.3}$$

(12 2019)

Solution: Evaluating the determinant of $\mathbf{A}$:

$$\left|\mathbf{A}\right| = 1\,(1 - 2) - 3\,(2 - 10) + 4\,(2 - 5) = 1\,(-1) - 3\,(-8) + 4\,(-3) = -1 + 24 - 12 = 1 \tag{1.2.34.4}$$

Using (1.1.32.1), computing the cofactor matrix and its transpose (adjugate):

$$\text{adj}(\mathbf{A}) = \begin{bmatrix} -1 & 1 & 2 \\ 8 & -19 & 6 \\ -3 & 14 & -5 \end{bmatrix} \tag{1.2.34.5}$$

Thus, $\mathbf{A}^{-1}$ is given by:

$$\mathbf{A}^{-1} = \frac{1}{11}\begin{bmatrix} -1 & 1 & 2 \\ 8 & -19 & 6 \\ -3 & 14 & -5 \end{bmatrix} \tag{1.2.34.6}$$

Writing the system of equations in matrix form $\mathbf{AX} = \mathbf{B}$ where $\mathbf{X} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 8 \\ 5 \\ 7 \end{pmatrix}$:

$$\mathbf{X} = \mathbf{A}^{-1}\mathbf{B} = \frac{1}{11}\begin{bmatrix} -1 & 1 & 2 \\ 8 & -19 & 6 \\ -3 & 14 & -5 \end{bmatrix}\begin{pmatrix} 8 \\ 5 \\ 7 \end{pmatrix} = \frac{1}{11}\begin{pmatrix} -8+5+14 \\ 64-95+42 \\ -24+70-35 \end{pmatrix} = \frac{1}{11}\begin{pmatrix} 11 \\ 11 \\ 11 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \tag{1.2.34.7}$$

Therefore, $x = 1$, $y = 1$, and $z = 1$.

1.2.35 Find the inverse of the matrix $\mathbf{A} = \begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$ using elementary transformations. (12 2019)

Solution: From (1.1.33.1), setting up the augmented matrix $[\mathbf{A} \mid \mathbf{I}]$ and applying Gauss-Jordan row elimination:

$$[\mathbf{A} \mid \mathbf{I}] = \left[\begin{array}{ccc|ccc} 2 & 0 & -1 & 1 & 0 & 0 \\ 5 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 3 & 0 & 0 & 1 \end{array}\right] \tag{1.2.35.1}$$

Performing elementary row transformations $R_2 \to R_2 - 2R_1$, row swaps, and elimination steps reduces the left block to the identity matrix $\mathbf{I}$, yielding:

$$\mathbf{A}^{-1} = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix} \tag{1.2.35.2}$$

1.2.36 A company produces two types of goods, A and B, that require gold and silver. Each unit of type A requires 3 g of silver and 1 g of gold while that of type B requires 1 g of silver and 2 g of gold. The company can use at most 9 g of silver and 8 g of gold. If each unit of type A brings a profit of Rs. 40 and that of type B Rs. 50, find the number of units of each type that the company should produce to maximize profit. Formulate the above LPP and solve it graphically and find the maximum profit. (12 2019)

Solution: Let x and y be the units of type A and type B goods produced. From

(1.1.34.1) and (1.1.34.2), the LPP formulation is:

$$\text{Maximize } Z = 40x + 50y \tag{1.2.36.1}$$
$$\text{subject to } 3x + y \leq 9 \tag{1.2.36.2}$$
$$x + 2y \leq 8 \tag{1.2.36.3}$$
$$x \geq 0,\ y \geq 0 \tag{1.2.36.4}$$

The feasible region is the polygon $OABC$ bounded by the extreme corner points:

$$O(0,0), \quad A(3,0), \quad B(2,3), \quad C(0,4) \tag{1.2.36.5}$$

Evaluating the objective function Z at each corner point:

- At $O(0,0) : Z = 0$
- At $A(3,0) : Z = 40\,(3) + 50\,(0) = 120$
- At $B(2,3) : Z = 40\,(2) + 50\,(3) = 80 + 150 = 230$
- At $C(0,4) : Z = 40\,(0) + 50\,(4) = 200$

By the Corner-Point Theorem, the maximum profit is Rs. 230, achieved by producing 2 units of type A and 3 units of type B. The following code verifies the graphical optimization:

codes/12/q36.py

1.2.37 There are three coins. One is a two-headed coin, another is a biased coin that comes up heads 75% of the time and the third is an unbiased coin. One of the three coins is chosen at random and tossed. If it shows heads, what is the probability that it is the two-headed coin? (12 2019)

Solution: Let E_1, E_2, E_3 be the events of choosing the two-headed coin, the biased coin, and the unbiased coin respectively.

$$\Pr(E_1) = \Pr(E_2) = \Pr(E_3) = \frac{1}{3} \tag{1.2.37.1}$$

Let A be the event that the tossed coin shows heads. The conditional probabilities are:

$$\Pr(A \mid E_1) = 1, \quad \Pr(A \mid E_2) = \frac{75}{100} = \frac{3}{4}, \quad \Pr(A \mid E_3) = \frac{1}{2} \tag{1.2.37.2}$$

Using Bayes' theorem (1.1.35.2):

$$\Pr(E_1 \mid A) = \frac{\Pr(E_1)\Pr(A \mid E_1)}{\Pr(E_1)\Pr(A \mid E_1) + \Pr(E_2)\Pr(A \mid E_2) + \Pr(E_3)\Pr(A \mid E_3)} \tag{1.2.37.3}$$

$$= \frac{\frac{1}{3}(1)}{\frac{1}{3}(1) + \frac{1}{3}\left(\frac{3}{4}\right) + \frac{1}{3}\left(\frac{1}{2}\right)} = \frac{1}{1 + \frac{3}{4} + \frac{1}{2}} = \frac{1}{\frac{9}{4}} = \frac{4}{9} \tag{1.2.37.4}$$

1.2.38 Find the equation of the line passing through $(2, -1, 2)$ and $(5, 3, 4)$ and of the plane passing through $(2, 0, 3)$, $(1, 1, 5)$ and $(3, 2, 4)$. Also, find their point of intersection. (12 2019)

Solution: The direction ratios of the line passing through $(2,-1,2)$ and $(5,3,4)$ are $3,4,2$. Its parametric Cartesian equation from (1.1.5.2) is:

$$\frac{x-2}{3}=\frac{y+1}{4}=\frac{z-2}{2}=\lambda \implies x=3\lambda+2,\ y=4\lambda-1,\ z=2\lambda+2 \tag{1.2.38.1}$$

The plane passing through $(2,0,3)$, $(1,1,5)$, and $(3,2,4)$ has normal $\mathbf{n}=\begin{pmatrix}-1\\1\\2\end{pmatrix}\times\begin{pmatrix}1\\2\\1\end{pmatrix}=\begin{pmatrix}-3\\3\\-3\end{pmatrix}$, yielding normal vector $\mathbf{n}_0=\begin{pmatrix}1\\-1\\1\end{pmatrix}$. From (1.1.28.1), the Cartesian equation of the plane is:

$$x-y+z=5 \tag{1.2.38.2}$$

From (1.1.37.1), substituting the line parameterization into the plane equation:

$$(3\lambda+2)-(4\lambda-1)+(2\lambda+2)=5 \implies \lambda+5=5 \implies \lambda=0 \tag{1.2.38.3}$$

Substituting $\lambda=0$ into the line equations yields the point of intersection:

$$\mathbf{P}=\begin{pmatrix}2\\-1\\2\end{pmatrix} \tag{1.2.38.4}$$

The following code verifies the intersection:

codes/12/q38.py